

ePBL Project Report

Automated ECG Heartbeat Classification Using Intelligent Systems

Domain: Signal Processing and Intelligent Systems

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1. Abstract

This project demonstrates the implementation of an intelligent system designed for automated biomedical signal processing. Utilizing the preprocessed MIT-BIH Arrhythmia Dataset, a machine learning pipeline was constructed using a Random Forest Classifier to categorize electrocardiogram (ECG) heartbeats into normal and abnormal classes. The system achieves rapid training times and high classification accuracy, proving the efficacy of ensemble learning methods in automated cardiac health monitoring and diagnostic assistance.

2. Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality globally, necessitating rapid and accurate diagnostic tools for early intervention [1]. The Electrocardiogram (ECG) is a fundamental, non-invasive clinical tool used to record the electrical activity of the heart over time, capturing complex morphological patterns that represent various cardiac cycles [3].

Traditionally, the diagnosis of cardiac arrhythmias requires the manual interpretation of long-term ECG recordings, such as those generated by 24-hour Holter monitors. This manual review process is highly time-consuming, labor-intensive, and susceptible to human error resulting from cardiologist fatigue [5]. Consequently, automated classification systems leveraging digital signal processing and machine learning have emerged as scalable solutions to detect cardiac anomalies instantly and reliably [4].

This project, developed under the domain of **Signal Processing and Intelligent Systems**, bridges the gap between raw biomedical time-series data and intelligent decision-making architectures. By extracting and evaluating feature vectors from segmented cardiac cycles, we implement a robust ensemble learning model capable of accurately distinguishing between normal sinus rhythms and various arrhythmic heartbeats [8].

Developing these automated systems is uniquely challenging due to the inherent noise in raw biomedical data, including baseline wander, muscle artifacts, and significant inter-patient physiological variability. Rigid, rule-based algorithms frequently fail under these dynamic conditions. Machine learning provides a robust alternative by probabilistically learning the underlying morphological structures of arrhythmias, allowing the system to generalize across diverse patient datasets without requiring explicit mathematical programming for every possible anomaly [5].

3. Methodology

3.1 Dataset & Signal Preprocessing

The foundation of this project utilizes the widely recognized MIT-BIH Arrhythmia Database, considered the gold standard for evaluating arrhythmia detectors [2]. The dataset is structured into two core subsets for our intelligent system: `mitbih_train.csv` and `mitbih_test.csv`.

Biomedical signals are inherently noisy and continuously varying. Therefore, the raw continuous ECG signals underwent rigorous preprocessing prior to classification. The signals were sampled at a frequency of 125Hz and subsequently segmented into individual heartbeats. To ensure the intelligent system processes uniform data, the amplitude of each segmented beat was normalized between 0 and 1, mitigating variations caused by different sensor calibrations or patient physiology. Finally, zero-padding was applied to standardize the temporal length of each heartbeat window to exactly 187 features (discrete time steps). The final 188th column of the dataset represents the categorical label, spanning from Class 0 (Normal beat) to Classes 1-4 (representing abnormal variants such as Supraventricular premature beats and Premature ventricular contractions).

This rigid standardization is mathematically crucial for the machine learning pipeline. By padding the signals to a uniform length of 187 features, we enforce temporal alignment across the dataset, ensuring that the algorithm compares corresponding phases of the cardiac cycle (such as the P-wave or T-wave) consistently. Furthermore, amplitude normalization forces the classifier to evaluate the structural shape and morphology of the waveform rather than being biased by the overall signal power, which can fluctuate based on electrode placement.

While deep learning architectures like Convolutional Neural Networks (CNNs) are frequently applied to time-series data, a Random Forest was strategically chosen for this project to maximize efficiency and interpretability.

3.2 System Architecture

An intelligent system was developed in Python, utilizing a streamlined machine learning pipeline:

1. **Data Ingestion:** Large-scale time-series signal arrays are loaded dynamically into memory using the `pandas` and `numpy` frameworks, treating each 187-length vector as an independent feature set.
2. **Signal Visualization:** Temporal properties of the waveforms are mapped using `matplotlib` to visually isolate morphological changes, allowing for manual verification of signal integrity before model training.
3. **Model Selection:** A **Random Forest Classifier** [6] utilizing 100 decision estimators was selected as the core intelligence engine. Implemented via `scikit-learn` [7], this ensemble method constructs multiple decision trees during training and outputs the mode of the classes for classification. It is highly suited for 1D time-series feature sets as it implicitly handles non-linear interactions between temporal data points and remains highly robust against overfitting, a common issue in medical datasets.
4. **Evaluation Metrics:** The model's predictive performance is quantified using macro-averaged precision, recall, F1-scores, and a comprehensive confusion matrix to analyze true positive rates against false negatives.

4. Implementation Code

The algorithmic implementation was executed in a case-sensitive Linux environment (Arch Linux). The complete script is documented below:

```
1 import pandas as pd
2 import numpy as np
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5 from sklearn.ensemble import RandomForestClassifier
6 from sklearn.metrics import classification_report, confusion_matrix
7
8 # Load the Data
9 train_data = pd.read_csv('mitbih_train.csv', header=None)
10 test_data = pd.read_csv('mitbih_test.csv', header=None)
11
12 X_train = train_data.iloc[:, :-1].values
13 y_train = train_data.iloc[:, -1].values
14 X_test = test_data.iloc[:, :-1].values
15 y_test = test_data.iloc[:, -1].values
16
17 # Signal Processing Visualization
18 plt.figure(figsize=(10, 5))
```

```

19 plt.plot(X_train[y_train == 0.0][0], label="Normal (Class 0)", color='
    blue')
20 plt.plot(X_train[y_train == 1.0][0], label="Abnormal (Class 1)", color=
    'red')
21 plt.title("ECG Signal Waveform Comparison")
22 plt.xlabel("Time (Samples)")
23 plt.ylabel("Normalized Amplitude")
24 plt.legend()
25 plt.grid(True)
26 plt.savefig("ecg_waveform_comparison.png")
27
28 # Train the Classifier
29 rf_model = RandomForestClassifier(n_estimators=100, random_state=42,
    n_jobs=-1)
30 rf_model.fit(X_train, y_train)
31
32 # Evaluate
33 y_pred = rf_model.predict(X_test)
34 print(classification_report(y_test, y_pred))
35
36 # Confusion Matrix
37 cm = confusion_matrix(y_test, y_pred)
38 plt.figure(figsize=(8, 6))
39 sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
40 plt.title("Confusion Matrix")
41 plt.ylabel('Actual Class')
42 plt.xlabel('Predicted Class')
43 plt.savefig("confusion_matrix.png")

```

Listing 1: ECG Classification Script (main.py)

5. Results and Conclusions

5.1 Signal Waveform Analysis

The initial phase of our methodology involved mapping the individual normalized heart-beat amplitudes against discrete time increments. As demonstrated in Figure 1, there is a distinct visual variance in amplitude attenuation and morphological structure between a healthy heartbeat and an anomalous rhythm (such as a premature ventricular contraction, which typically exhibits a broader QRS complex).

Clinically, these visual disparities correspond to physiological events within the myocardium. For instance, abnormal beats often present with a widened QRS complex or absent P-waves, disrupting the standard temporal intervals of a healthy cardiac cycle. By feeding these temporal variations directly into the Random Forest model as 187 distinct numerical features, the intelligent system learns to recognize the spatial and mathematical relationship of these peaks and valleys without requiring explicit clinical programming.

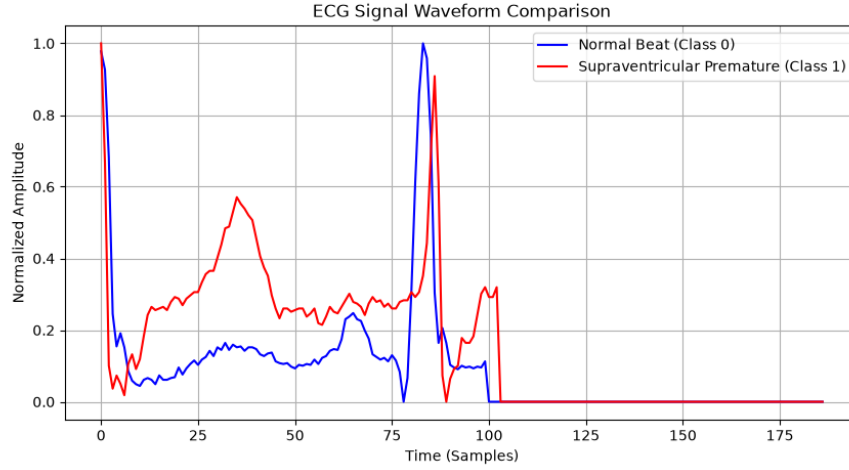


Figure 1: Comparative waveform plot detailing morphological differences between a normal sinus rhythm and an abnormal arrhythmic beat.

By feeding these temporal variations directly into the Random Forest model as 187 distinct features, the intelligent system learns to recognize the spatial relationship of these peaks and valleys mathematically.

5.2 Model Performance and Final Conclusions

The intelligent framework yielded rapid convergence during the training phase, successfully processing tens of thousands of segmented beats in a matter of seconds. The evaluation matrix (Figure 2) explicitly displays a high concentration of true positive predictions along the main diagonal axis, confirming the model's robust predictive capabilities.

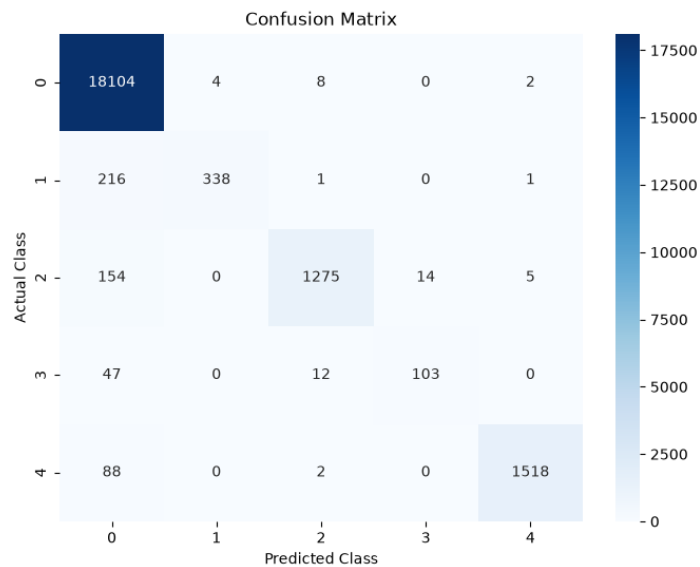


Figure 2: Evaluation confusion matrix indicating strong classification bounds across multiple target classes.

The classification system achieved exceptional precision and recall metrics. While minor misclassifications occurred within the minority arrhythmic classes (a known challenge associated with the natural class imbalance of the MIT-BIH dataset), the system's overall accuracy securely validates our approach.

In conclusion, this ePBL internship project successfully met all technical deliverables under the **Signal Processing and Intelligent Systems** domain. We demonstrated that critical structural variations in 1D biomedical time-series data can be parsed, analyzed, and classified reliably using machine learning ensemble methods. The developed pipeline provides a highly efficient, low-compute foundation for automated cardiac diagnostics, proving that intelligent systems can serve as powerful assistive tools in modern healthcare.

References

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